JEANNETTE BASSÈNE

Engineering Student - Supmicrotech ENSMM

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FINAL-YEAR INTERNSHIP: 6 months, from Feb 2027

PROFILE

Engineering student at Supmicrotech ENSMM, specializing in connected objects. Cross-disciplinary profile (electronics, embedded systems, mechanics, computer science) with hands-on experience in instrumentation, simulation and experimentation.

EXPERIENCES

Internship - Cristal Innov

Sept 2025 – Feb 2026 | Instrumentation of a hydrothermal quartz synthesis autoclave

- Led a feasibility study on non-destructive monitoring (MATLAB, in-situ tests), successfully delivering viable results that characterized system boundaries and validated recommendations for future development.
- Contributed to production and quality operations (precision machining, IR spectroscopy, optical profilometry, 8D problem solving).

Private Tutoring - ACADOMIA

Student job, since Sept 2026

Personalized tutoring in Mathematics, Physics and Chemistry.

ACADEMIC PROJECTS

OBCP Prototyping (L476RG) – Oct 2026

Executed end-to-end manufacturing of an IoT board: solder paste deposition, manual and automated component placement (Kaio, SMTech), reflow soldering, microscopic quality control, electrical testing, and final programming.

Homodyne Michelson Interferometry – Mar 2026

Performed optical vibration measurements using a homodyne Michelson interferometer; processed experimental signals in MATLAB to extract and interpret vibration data.

Ultrasound Simulation (MATLAB) – June 2025

Developed a multilayer acoustic model for ultrasound propagation; estimated time-of-flight using cross-correlation and evaluated signal-to-noise ratio (SNR) to assess robustness to noise.

Crowd Simulation (Java) – June 2025

Developed an object-oriented simulation of autonomous agents navigating toward an exit, implementing collision detection, collision avoidance and obstacle avoidance; tested agent behaviour in evacuation scenarios.

EDUCATION

Supmicrotech ENSMM (2024 – 2027)

Engineering degree (Master's level) – Specialization in Design and Development of Connected Objects.

Lycée Henri Poincaré, Nancy, France (2022 – 2024)

Two-year intensive undergraduate program in Physics, Chemistry, and Engineering Sciences (PCSI/PC*).

High School Diourbel, Senegal (2021 – 2022)

Scientific Baccalaureate: Mathematics, Physical Sciences (High Honors - Scientific excellence high school).

SKILLS

Programming: MATLAB, Python, C, SQL, LabVIEW (Proficient).
Mechanics & CAD: Mechanical Design & CAD (Working knowledge).
Management: Technical Communication, Project Management.

LANGUAGES, SPORTS & INTERESTS

Languages: English (C1 - Linguaskill test), French (Bilingual), Italian (A2), Spanish (B2), Wolof (Native language).
Sports: Running.
Interests: Playing the flute, Programming, Logic games.